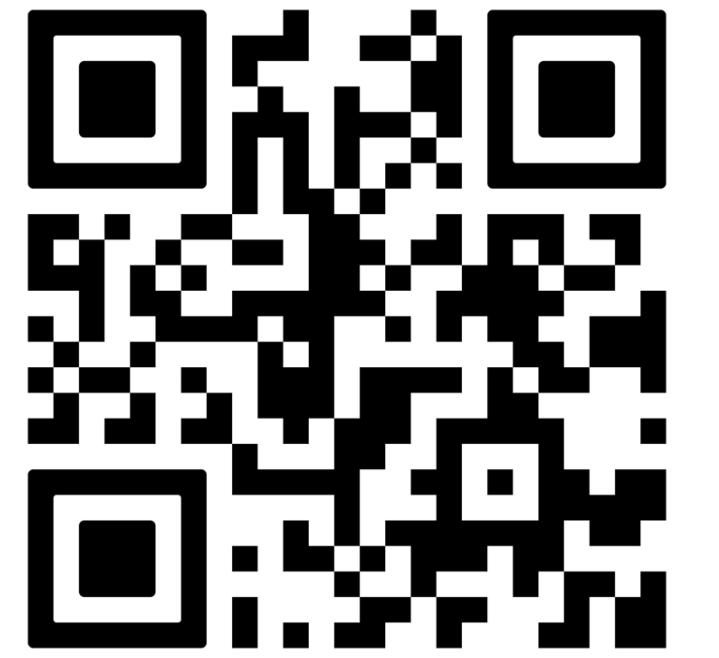


# Learning Holistic Whole-Body Loco-Manipulation with a Bipedal Mobile Manipulator

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\* Equal contribution



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# INTRODUCTION

## MOTIVATION

Manipulation targets often lie beyond the nominal workspace of a fixed-base arm. A bipedal platform can extend its reach through coordinated arm motion, postural adaptation, and stepping. Yet many systems require separate base, gait, or body-posture commands, forcing high-level policies to specify how the whole body should move.

*Can a single instantaneous 6-DoF end-effector target serve as the sole task command? The controller must decide when arm motion is sufficient and when to recruit the lower body, while jointly handling tracking, locomotion, and dynamic balance under partial observations.*

## CONTRIBUTIONS

- **End-Effector-Only Whole-Body Control:** We formulate bipedal loco-manipulation using a single instantaneous 6-DoF target to drive all 14 arm and leg joints, enabling autonomous reaching, postural adaptation, and stepping without explicit locomotion commands.
- **Unified Learning Framework:** We combine phase-aware reward gating for tracking-locomotion-balance coordination with a Transformer-GRU temporal context estimator; simulation ablations verify the contribution of both components.
- **Reusable Real-World Interface:** The same deployed controller executes commands from Quest teleoperation, a UMI-based diffusion policy, and scripted trajectories, while expanding vertical reach from 38–163 cm to 3–191 cm.

## METHOD

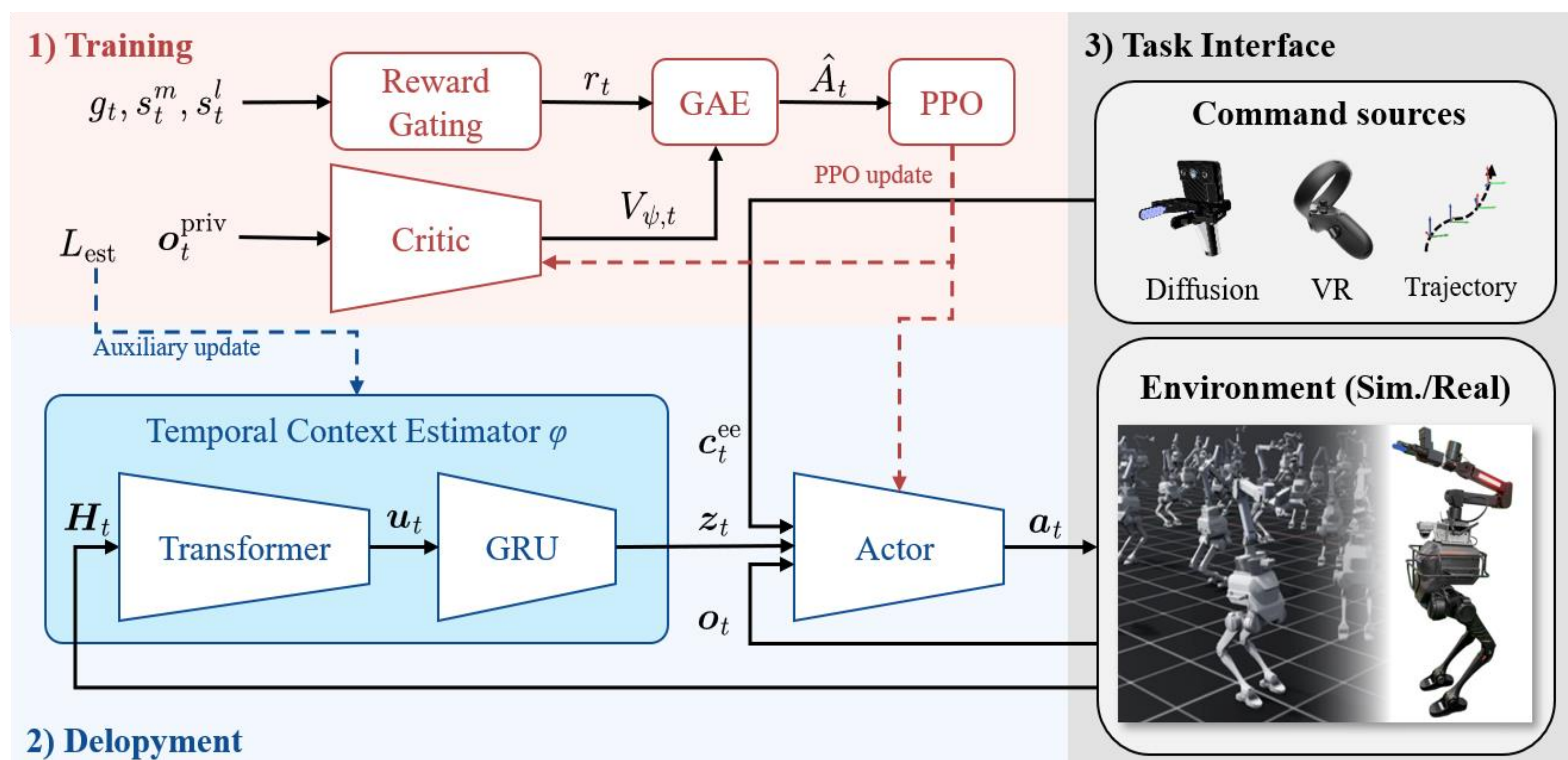


Fig. 1. Unified framework for reward-gated training, temporal context estimation, and deployment through a common 6-DoF task interface.

## CORE REWARD TERMS

| Term                 | Definition                                                   | Scale |
|----------------------|--------------------------------------------------------------|-------|
| Position tracking    | $r_t^p = \exp(-e_t^p/\sigma_p^2) + \exp(-5e_t^p/\sigma_p^2)$ | 6.0   |
| Coarse orientation   | $r_t^{R,C} = \exp[-(e_t^{R/1.5})^2]$                         | 6.0   |
| Fine orientation     | $r_t^{R,f} = \exp(-e_t^p/0.5) \exp[-(e_t^R/0.25)^2]$         | 6.0   |
| Reference tracking   | $r_{ref} = \exp[-e_{ref} / (\sigma_{ref})^2]$                | 10.0  |
| Best-so-far progress | $r_t^{prog}$                                                 | 15.0  |
| Phase safety         | $(1 - g_s) s_t^m + g_s s_t^l$                                | 2.0   |

## PHASE-AWARE REWARD GATING

$$g_t = \text{sigmoid}(k(d_t^{\text{ref}} - \mu))$$

$$\tilde{\mathbf{S}}_t^\varphi = \mathbf{S}_t^\varphi + 0.4$$

- Large  $g_t$ : locomotion and target approach.
- Small  $g_t$ : precise tracking and stabilization.
- Relaxed gates retain recovery learning signals.

## TEMPORAL CONTEXT

- Transformer: dependencies within the latest observation window.
- GRU: information beyond the fixed window.
- Auxiliary prediction trains the estimator; deployment uses onboard observations only.

## EXPERIMENTS &

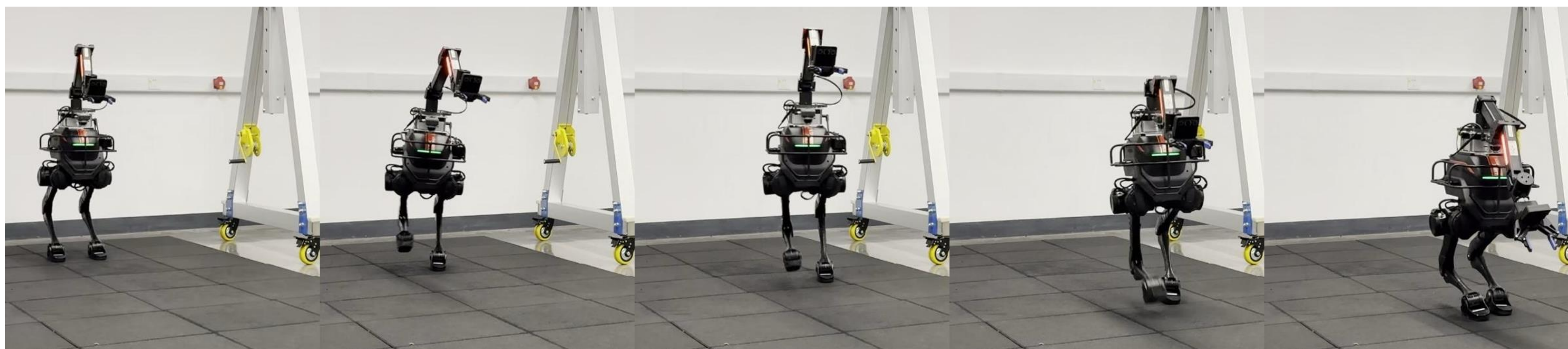


Fig. 2. Successive frames of autonomous whole-body adaptation during end-effector tracking.

## ABLATION RESULTS

| Method                         | Success $\uparrow$ | Pos. Mean $\downarrow$ | Action Var. $\downarrow$ |
|--------------------------------|--------------------|------------------------|--------------------------|
| Matched additive               | 82.73%             | 3.23 cm                | 0.218                    |
| No latent                      | 69.87%             | 3.65 cm                | 0.236                    |
| Transformer only               | 84.43%             | 3.21 cm                | 0.160                    |
| <b>Ours: Transformer + GRU</b> | <b>88.30%</b>      | <b>2.85 cm</b>         | <b>0.165</b>             |
| Privileged oracle              | 94.53%             | 2.51 cm                | 0.112                    |

## SIMULATION RESULTS

- **Reward gating** raises success from **82.73% to 88.30%**, reduces mean position error from **3.23 to 2.85 cm**, and lowers action variation from 0.218 to 0.165.
- **Temporal context** is essential: removing it drops success to **69.87%**; Transformer-GRU gives the best non-oracle result.

## REAL-WORLD DEMONSTRATIONS

- **One controller, three command sources:** Quest teleoperation, a UMI-based diffusion policy, and scripted trajectories—without explicit base, gait, or footstep commands.
- **Demonstrated tasks:** pickup-and-place with stepping, whiteboard wiping, cabinet closing, and vertical tracking.
- Reachable height expands from **38–163 cm** to **3–191 cm**.

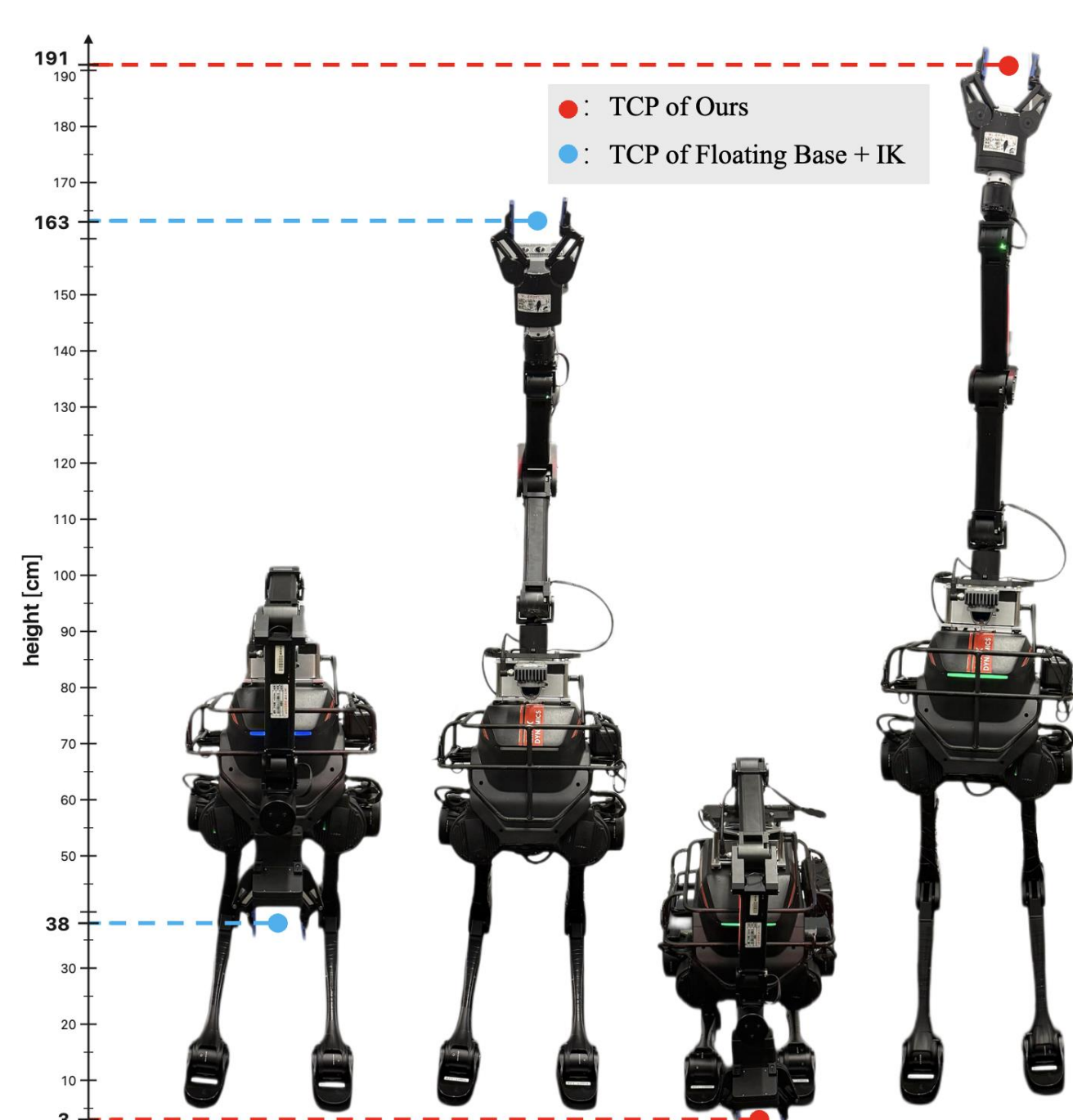


Fig. 3. Vertical workspace comparison. Our controller expands the reachable height from 38–163 cm to 3–191 cm.

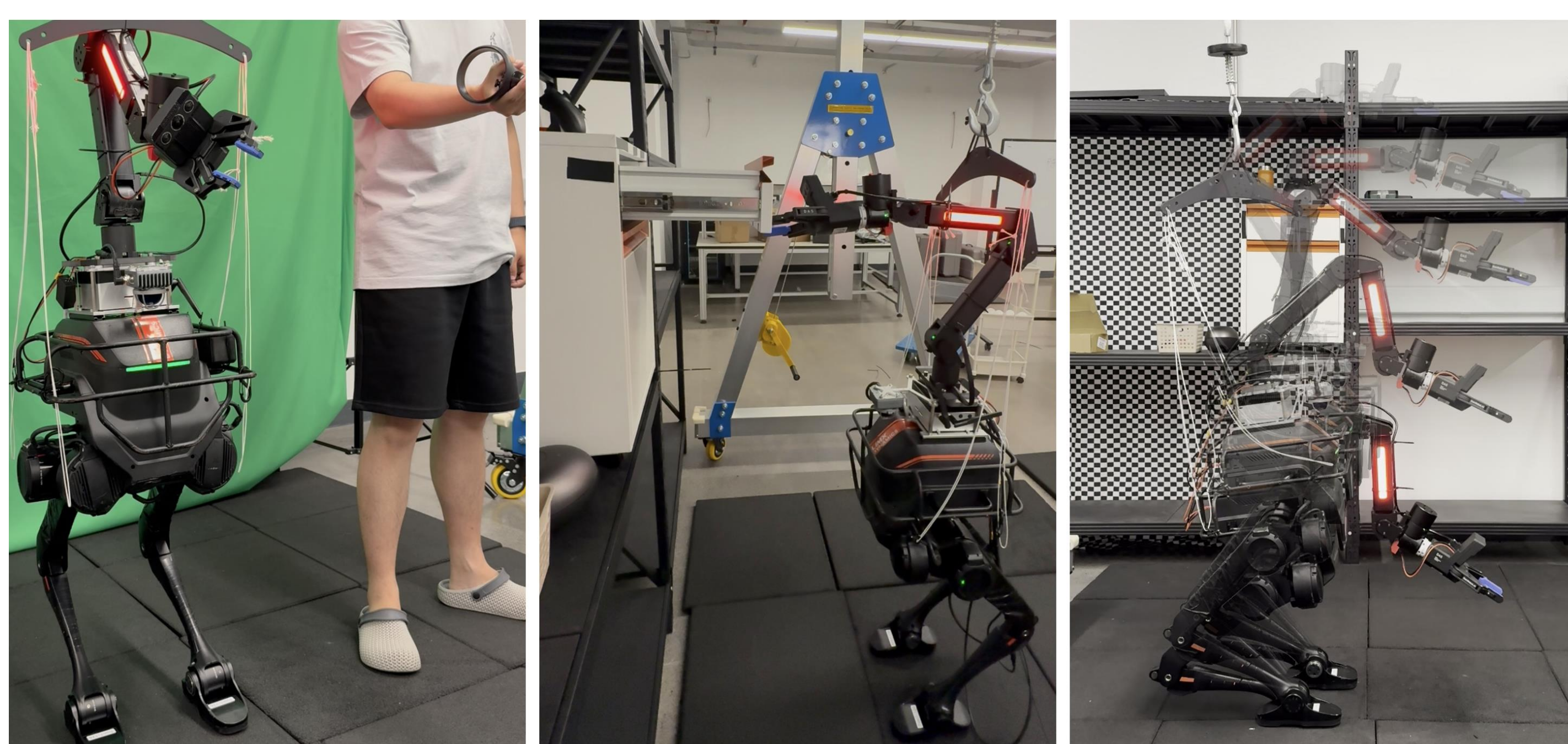
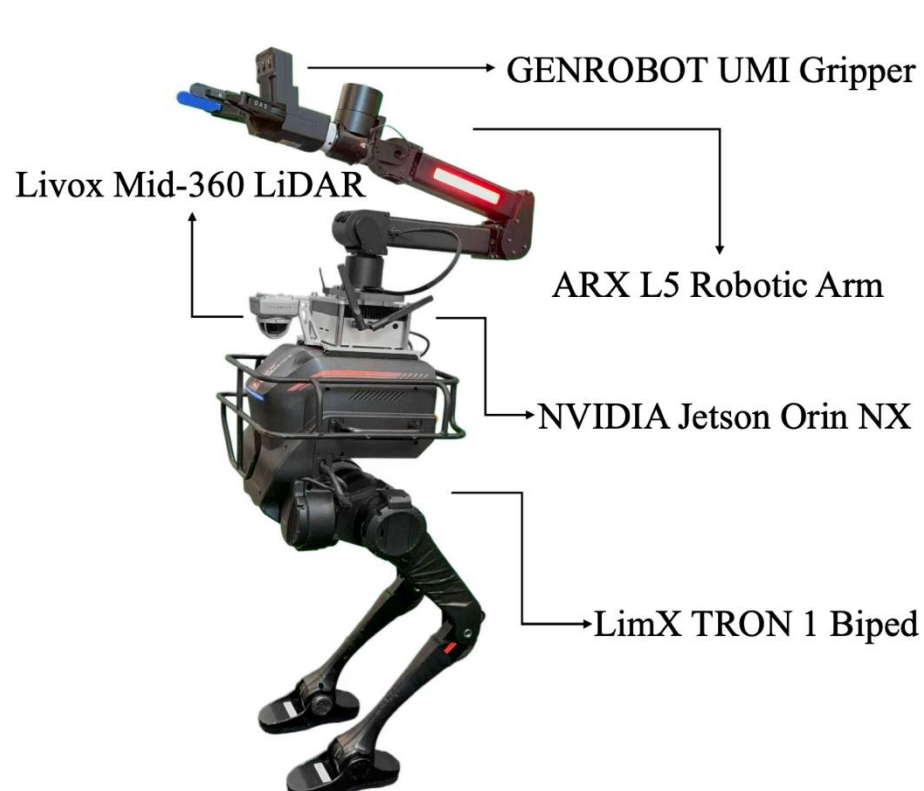


Fig. 4. Unified real-world interface for Quest teleoperation, a UMI-based diffusion policy, and scripted trajectory tracking.

## CONCLUSION & OUTLOOK



## CONCLUSION

- **Unified whole-body control:** A single instantaneous 6-DoF end-effector target drives all 14 arm and leg joints without explicit base-velocity, gait, or footstep commands.
- **Coordinated learning:** Reward-gated training balances tracking, locomotion, and stability, while Transformer-GRU temporal context supports deployment under partial observations.
- **Simulation and real-world validation:** The controller autonomously reaches, adapts posture, squats, and steps; it also provides one interface for human, learned, and scripted commands.

## FUTURE WORK

- **More dynamic loco-manipulation:** Extend beyond the current reaching-centric behaviors toward faster whole-body transitions, moving targets, and more dynamic manipulation.
- **Finer task-space tracking:** Improve end-effector precision for fine-grained manipulation, particularly when following learned high-level commands.
- **Broader validation:** Evaluate longer-horizon tasks, stronger disturbances, and additional bipedal mobile-manipulator platforms.